System Design

Surette Nexus · Surette Real Estate, Waterville, Maine

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How Surette Nexus is built to stay fast, correct and observable as it grows: from one laptop, to a small Docker host, to a Kubernetes cluster. Everything here is free/open-source software; nothing requires a paid service.

> **What's verified.** The application code (rate limiting, caching, idempotency, circuit breakers, queue, outbox, WebSockets, health checks, logs, metrics) is covered by automated tests. The full Docker Compose stack (two web instances, worker, WebSocket server, Postgres, Valkey, nginx gateway) has been built and run locally, and the browser tests pass through the gateway. Alert rules, Alertmanager routing and Kubernetes manifests pass their official validators, and the GitHub Actions CI pipeline passes on GitHub. Not yet run: a real Kubernetes cluster, the Prometheus/Grafana/Loki containers, and the deploy workflow (they need hosting). See Verification status.

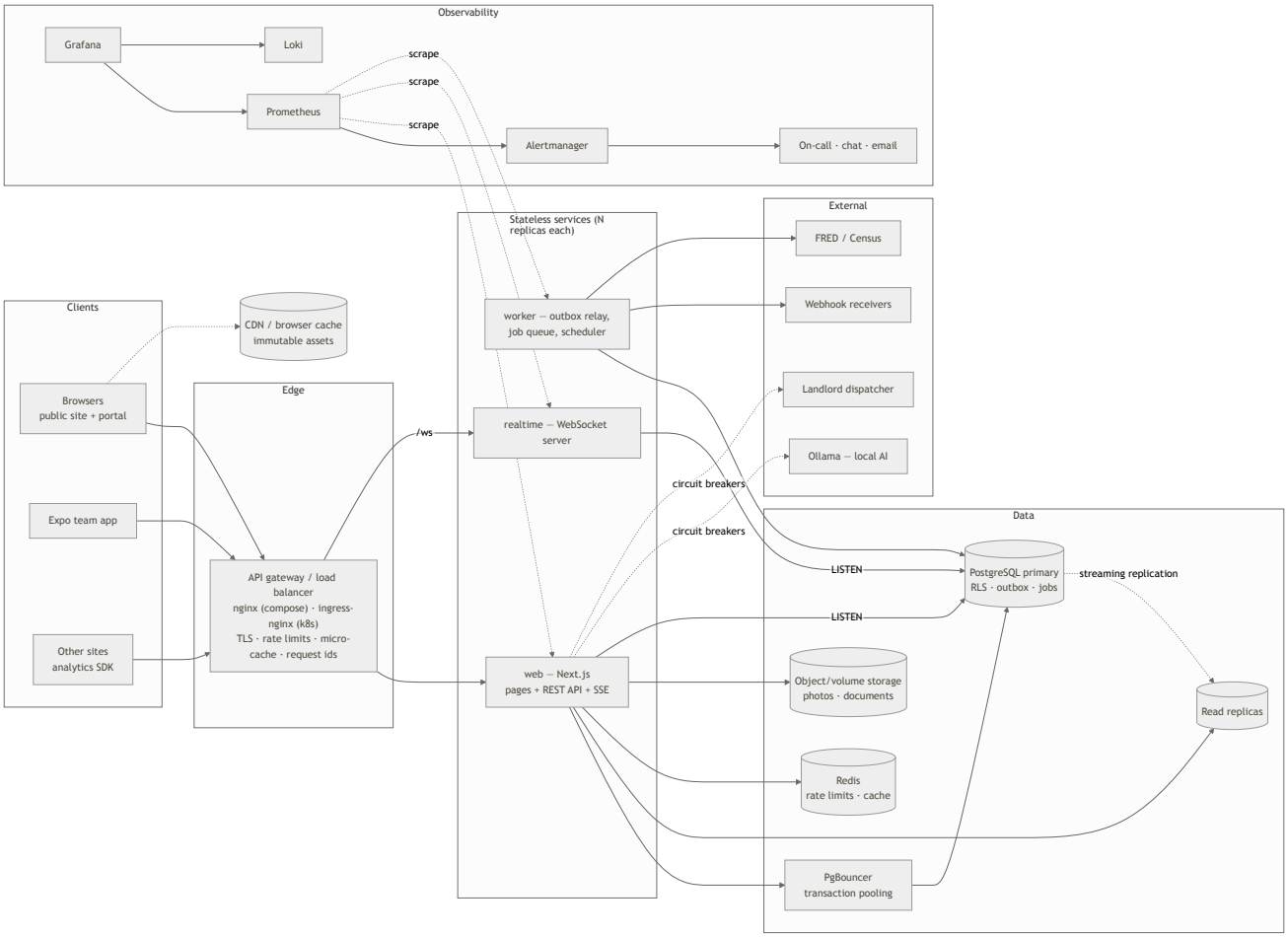
1. Goals and constraints

Goal	What it means here
Correct first	Money in integer cents, PostgreSQL row-level security on every request, no duplicate leads or deals from retries, no lost events
Fast for visitors	Public pages and listing APIs answer from cache; images are pre-sized WebP; live updates are pushed instead of polled
Degrade, don't fail	If AI, FRED, Census, the dispatcher, Redis or a webhook receiver is down, the site keeps working with fallbacks

Goal	What it means here
Scale horizontally	Every process is stateless; shared state lives in Postgres (and optionally Redis); any number of replicas
Observable	Every request has an id, a structured log line and metrics; every alert has a runbook
Free / OSS	Postgres, Valkey (Redis-compatible), nginx, Prometheus, Grafana, Loki, Alertmanager, KEDA, CloudNativePG, Ollama

Expected load is modest (a regional brokerage: hundreds to low thousands of visitors a day), so the design favours **few moving parts now** with a **clear path** to more when numbers justify it.

2. Architecture

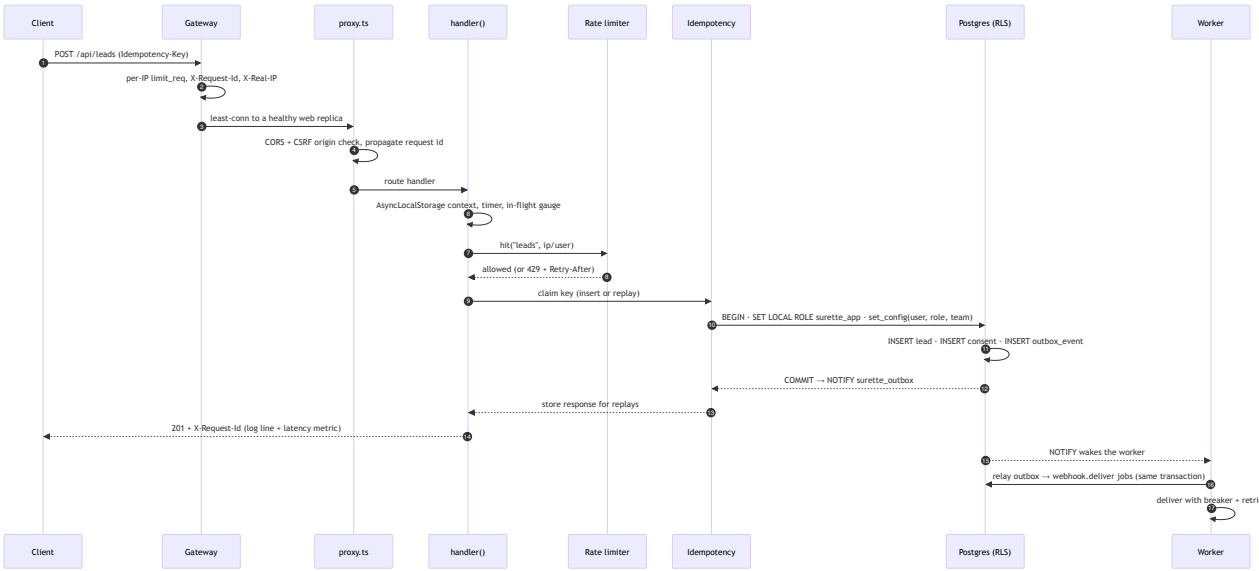


Component	Responsibility	State	How it scales
web (Next.js)	Pages, REST API, SSE stream, request auth, RLS-scoped queries	none (L1 cache only)	HPA on CPU/memory, 3 to 20 replicas
realtime	WebSocket live updates	open sockets only	HPA on CPU; each replica LISTENs to Postgres directly

Component	Responsibility	State	How it scales
worker	Outbox relay, job processing, scheduled jobs	none	KEDA on queue depth, 2 to 10 replicas; SKIP LOCKED makes them safe to run in parallel
gateway	TLS, routing, rate limits, micro-cache, request ids	cache (disposable)	Cloud LB / ingress controller replicas
PostgreSQL	Source of truth, RLS, outbox, queue, idempotency keys	durable	Vertical first; read replicas; partitioning for analytics
PgBouncer	Connection pooling (transaction mode)	none	2 replicas
Redis (optional)	Shared rate-limit counters, shared cache, invalidation pub/sub	disposable	Single node is plenty; managed Redis for HA

Code map: `web/src/lib/server/*` (one module per concern, named below), `web/scripts/worker.ts`, `web/scripts/realtime.ts`, `deploy/` (gateway, Kubernetes, observability), `.github/workflows/`.

3. Life of a request



4. API gateway and load balancing

Where: `deploy/nginx/nginx.conf` (Docker Compose) and `deploy/k8s/base/ingress.yaml` (ingress-nginx).

- **Routing:** `/ws` goes to *realtime* (WebSocket upgrade, 1 h timeouts); `/api/events` goes to *web* with buffering off (SSE); `/api/properties/upload` streams bodies up to 1.2 GB with no request

buffering; everything else goes to *web*.

- **Load balancing:** `least_conn` across replicas, upstream keep-alive, passive health checks (`max_fails=3 fail_timeout=10s`). In Kubernetes, readiness probes remove unhealthy pods from the Service.
- **Retries:** another replica is tried only on connection errors and 502/503, and **only for idempotent methods**. nginx never retries POST/PATCH unless told to with `non_idempotent`, and we don't. Clients retry writes safely with `Idempotency-Key` (§8).
- **Identity:** the gateway overwrites `X-Real-IP` with the address it saw (the app trusts only this header for per-IP limits) and sets or keeps `X-Request-Id`.
- **Protection:** `/api/metrics` returns 404 at the nginx gateway. Under ingress-nginx it is reachable but needs `METRICS_TOKEN` in production, and Prometheus scrapes pods on the private network. The app adds security headers itself, so they are the same behind any gateway.

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5. Rate limiting

Two layers:

Layer	Where	Key	Purpose
Coarse	Gateway (`limit_req`)	client IP	Absorb floods before they reach Node: 20 r/s for the API, 10 r/min for login and sign-up, 10 r/s for analytics
Precise	App (`lib/server/limiter.ts`)	endpoint + user id (signed in) or IP	Business limits, shared across all replicas

App limits (fixed window): login 20 per 15 min, sign-up 10 per hour, contact/leads 6 per hour, showings 8 per hour, AI assistant 30 per 10 min, ops assistant 60 per minute, Dual-Move share 5 per hour, analytics 240 per minute, maintenance 5 per 15 min.

- **Store:** Redis (`INCR` + `PEXPIRE`) when `REDIS_URL` is set; otherwise an UNLOGGED Postgres table (`rate_limit_counters`, one upsert per hit), so limits stay shared across replicas without Redis. In-memory is for development only.
- **Responses:** `429` with `Retry-After` and `RateLimit-Limit/Remaining/Reset`.
- **Fail open:** if the shared store is unreachable, limits fall back to per-process memory. A limiter outage must not take the site down. Account lockout (8 failures, then 15 minutes) still protects logins.
- **Why fixed window:** simple and cheap, and the bursts at window edges are small at these limits. A sliding-window log is a drop-in change in `limiter.ts` if needed.

6. Caching

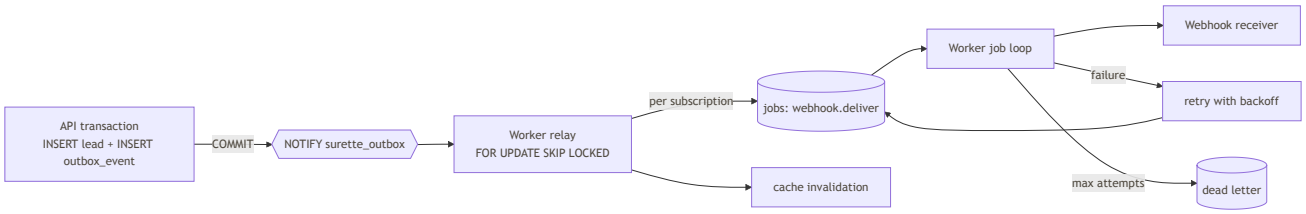
Layer	What	TTL / invalidation
Browser/CDN	`/_next/static/*`, `/media/*` (content-hashed WebP renditions)	1 year, immutable
Gateway micro-cache	Anonymous `GET /api/properties`, `/api/market`	`s-maxage` from the app (15 s / 10 min) + stale-while-revalidate; bypassed when a session cookie or Authorization header is present; `proxy_cache_lock` collapses concurrent misses
App L1 (in-process LRU, 500 entries)	Public listings, cities, team, team directory, market series	60 s / 5 min / 10 min; **dropped instantly** when the Postgres change feed reports a listing or photo change
App L2 (Redis, optional)	Same keys, shared across replicas	Tag sets (`cachetag:listings`), deleted on change; pub/sub tells other replicas to drop L1
Database	Shared buffers, indexes	—

Rules: **only anonymous/public data is cached.** Anything that depends on who is signed in always reads through RLS. Concurrent misses for a key share one load (request coalescing), so a popular entry expiring doesn't cause a stampede. Code: `lib/server/cache.ts`, `lib/server/live-cache.ts`.

7. Event-driven architecture and the message queue

Transactional outbox

Business code records domain events **in the same transaction** as the change (`publishEvent(db, …)` in `lib/server/events.ts`). An event exists if and only if the change committed. There is no "saved the lead but lost the notification" (dual-write) bug.



Event catalog: `lead.created`, `deal.created`, `deal.stage_changed`, `showing.requested`, `showing.updated`, `listing.published`, `listing.updated`, `media.uploaded`, `collection.activity`.

Job queue

A durable queue on PostgreSQL (`lib/server/jobs.ts`, table `jobs`):

- Workers **claim with** `FOR UPDATE SKIP LOCKED`, so any number of replicas can pull from the same queue without double-processing (tested).
- **Retries:** exponential backoff with jitter (5 s, 20 s, 80 s, … capped at 1 h). After `max_attempts` a job is marked `dead`; that alerts and links to its runbook, and the job can be re-queued with one SQL statement.
- **Visibility timeout:** jobs stuck in `running` (a crashed worker) are released after 5 minutes.
- **Dedupe keys:** at most one queued or running job per key (for example one market refresh per day).
- **Wake-ups:** `NOTIFY surette_jobs` starts work within milliseconds; polling every 2 s is the safety net.
- **Queues:** `default`, `webhooks` and `maintenance` keep slow webhook receivers from starving other work.
- **Emails** (`email.send`, 6 attempts): confirmation and password-reset emails are queued in the same transaction that creates the token, so a link is never emailed for a rolled-back sign-up, and a mail server outage only delays delivery.
- **Scheduler:** the worker holding a Postgres advisory lock enqueues recurring jobs (hourly cleanup, daily analytics purge, daily market refresh, monthly government data refresh). Per-period dedupe keys make a split brain harmless.

Delivery semantics: at-least-once. Consumers are idempotent: webhook deliveries carry a stable `X-Surette-Delivery` id, and job dedupe keys block duplicate enqueues.

Why Postgres instead of Kafka/RabbitMQ (for now): it's already there, it's transactional with the business data (no outbox-to-broker gap), and it easily handles thousands of jobs per minute. Switch when sustained throughput reaches tens of thousands of messages per second, or when many independent consumers need replay. The outbox then feeds the broker (Debezium CDC or the relay publishing to NATS JetStream or RabbitMQ) without changing business code.

8. Idempotency

Mechanism	Where	Protects against
`Idempotency-Key` header	`POST /api/leads`, `/api/showings`, `/api/deals`, `/api/deals/:id/drafts`, `/api/deals/:id/documents`, `/api/properties/upload`, `/api/dual-move/share`	Double-clicks, mobile retries, gateway timeouts creating duplicates
Analytics `event_id` (UUID per event)	`/api/analytics/collect`	SDK resending a batch (`ON CONFLICT DO NOTHING`)
Webhook delivery id	worker	Receivers processing a retried delivery twice

Mechanism	Where	Protects against
Job dedupe keys	queue	Duplicate scheduled jobs and fan-out
Business duplicate rule	leads	Same person and property within 24 h becomes a note, not a new lead

Contract (`lib/server/idempotency.ts`): keys are 8 to 128 characters, scoped per user (or per IP when anonymous), and kept for 24 h.

- **Repeat:** returns the stored response with `Idempotent-Replayed: true`.
- **Same key, different body:** `422`.
- **Repeat while the original is still running:** `409`.
- **Failure:** 5xx responses and exceptions release the key, so the client can retry.

A take-over of an expired key is a single atomic upsert, so two simultaneous first requests can't both run. JSON bodies are hashed. Multipart uploads stream straight to storage, so they are keyed on content type and length instead.

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9. Resilience: circuit breakers, timeouts, retries

`lib/server/circuit-breaker.ts` has three states. *Closed* counts failures. *Open* fails fast for a cool-down. *Half-open* lets one trial call through. Every call also gets a timeout.

Dependency	Breaker	Threshold / cool-down / timeout	Fallback while open
Ollama (local AI)	`ollama`	3 / 30 s / 12 s	Deterministic built-in answers (always correct, less chatty)
FRED	`fred`	5 / 5 min / 20 s	Last stored observations, shown with their date
Census, BLS, FHFA downloads	`gov-data`	6 / 5 min / 120 s	Previous government data stays; the error is recorded on the series
Census ACS	`census`	5 / 5 min / 20 s	Same
Landlord dispatcher	`dispatcher`	4 / 30 s / 12 s	"Temporarily unavailable — call the office" (4xx answers don't count as failures)
Each webhook host	`webhook:<host>`	5 / 60 s / 8 s	Job retried later with backoff
SMTP (account emails)	`smtp`	5 / 60 s / 20 s	Email job retried later with backoff

Dependency	Breaker	Threshold / cool-down / timeout	Fallback while open
Google sign-in token exchange	`google`	5 / 60 s / 10 s	Customer sees "Google sign-in didn't complete" and can use email

Other patterns:

- **Bulkheads:** background work runs in a separate process, and webhooks have their own queue.
- **Statement timeout:** 15 s per database statement, so a runaway query can't hold a connection forever.
- **Graceful shutdown:** web pods sleep 10 s in `preStop` so the load balancer drains them. Workers stop claiming and finish in-flight jobs for up to 25 s. The realtime server closes sockets with code 1012, and clients reconnect to another replica with backoff and jitter.

10. Real-time: WebSockets and SSE

Database triggers `NOTIFY surette_events` on every change to listings, photos, showings, collections, leads and deals, no matter who made the change. Both transports apply the same visibility rules (`lib/server/live-filter.ts`) and send only ids. Clients re-fetch through the API, where RLS applies.

	WebSocket (`/ws`, `realtime` service)	SSE (`/api/events`, web)
Used by	Browsers when `NEXT_PUBLIC_REALTIME_URL` is set; the mobile app can use it too	Fallback after 3 failed WebSocket attempts, or when not configured
Auth	Session cookie (Origin must be our site, which blocks cross-site WebSocket hijacking) or a `bearer.<token>` subprotocol for native apps, so tokens never appear in URLs or logs	Session cookie
Liveness	Protocol ping every 25 s; dead peers terminated	Comment every 20 s
Protection	Max 20 connections per IP, 4 KB frames, slow consumers dropped at 1 MB buffered	Gateway connection limit

Scaling: each replica holds one LISTEN connection to Postgres, so there is no shared broker and replicas scale independently. LISTEN needs a session connection, so it uses `DATABASE_DIRECT_URL` and bypasses PgBouncer. At very high fan-out (many thousands of sockets per replica), add Redis pub/sub between the database listener and socket servers.

11. Database: indexing and scaling

Indexes (migrations 001, 002 and 005) follow the actual query patterns:

Query	Index
Public listing grid (status, featured, newest)	`properties_public_idx (status, is_featured DESC, published_at DESC)`
Filters (type, beds, price)	`properties_filter_idx`, partial `properties_price_idx … WHERE status='active'`
City filter `lower(city) = lower($1)`	expression index `properties_city_idx (lower(city), status)`
AI assistant fuzzy town match	trigram GIN `properties_city_trgm_idx`
Keyword search	GIN on `search_vector`
Deadline agenda (open items by due date)	partial `… WHERE completed_at IS NULL` / `WHERE status='open'`
Lead duplicate check (email/phone within 24 h)	partial `leads_email_idx`, `leads_phone_idx`
Follow-ups due	partial `leads_follow_up_idx`
Job claiming	partial `jobs_ready_idx (queue, priority DESC, run_at, id) WHERE status='queued'`
Outbox relay	partial `outbox_unpublished_idx WHERE published_at IS NULL`
Analytics time ranges (append-only, huge)	**BRIN** on `occurred_at` (tiny, fast for ranges) + B-tree for project/type/session

Connections: each web replica has a pool (`DATABASE_POOL_SIZE`, default 10) behind PgBouncer in **transaction mode**. This is safe because every request is one transaction and identity uses `SET LOCAL` / `set_config(…, true)`.

Reads: `withReadDb` and `withSystemReadDb` run read-only transactions on `DATABASE_READ_URL` replicas, for analytics dashboards, market data and exports. Never read your own writes from a replica.

Growth path, in order, each step taken only when metrics call for it:

1. Vertical scaling plus `pg_stat_statements`-driven indexing.
2. Read replicas (CloudNativePG gives `-ro` automatically).
3. Partition `analytics_events` by month (`PARTITION BY RANGE (occurred_at)`) once it passes about 50 M rows. Retention then becomes `DROP PARTITION` instead of `DELETE`.
4. Move photos and documents to S3/R2 behind a CDN, which removes the shared volume.
5. Separate the analytics store (ClickHouse, OSS) if dashboard queries start competing with transactional load.

Durability: CloudNativePG runs 1 primary and 2 replicas with automatic failover, continuous WAL archiving and nightly base backups (30-day point-in-time recovery). Migrations are forward-only and **expand → migrate → contract**, so old and new pods can run side by side during a rolling deploy.

12. Health checks

Endpoint	Probe	Checks	On failure
`/api/health/live`	Kubernetes liveness + startup, Docker HEALTHCHECK	Process answers HTTP, nothing external	Container restarted
`/api/health/ready`	Kubernetes readiness, blackbox exporter	Database reachable, all migrations applied, pool not exhausted, Redis (if configured)	Pod removed from the load balancer until it recovers
`/api/health`	Humans, uptime monitors	Same as ready, plus circuit-breaker states	—
worker `:9464/healthz`, `/readyz`	Kubernetes	Job loop alive, database, LISTEN	Restart / unready
realtime `:3200/healthz`, `/readyz`	Kubernetes	Not draining	Restart / unready

Liveness never checks dependencies, so a database blip doesn't restart every pod at once.

13. Observability

Logs (`lib/server/logger.ts`, pino): one JSON object per line with `service`, `component`, `requestId`, `route` and `userId`. Passwords, tokens, cookies, emails and phone numbers are redacted. Each API request logs method, route, status and duration; requests slower than 2 s are `warn`, and 5xx responses are `error`. Promtail ships container logs to **Loki**; query them in Grafana with `{service="surette-web"} | json | requestId="…"`.

Metrics (`lib/server/metrics.ts`, prom-client) at `/api/metrics` (token-protected), worker `:9464/metrics` and realtime `/metrics`:

Metric	Type	Labels
`http_request_duration_seconds`	histogram	method, route (ids collapsed), status
`http_requests_in_flight`	gauge	—
`db_transaction_duration_seconds`	histogram	kind (rls, system, read)
`db_pool_connections`	gauge	state (total, idle, waiting)
`rate_limited_total`	counter	bucket
`idempotent_replays_total`	counter	route
`cache_requests_total`	counter	cache (l1, l2), result

Metric	Type	Labels
`circuit_breaker_state`, `circuit_breaker_rejections_total`	gauge / counter	breaker
`external_call_duration_seconds`	histogram	target, outcome
`jobs_processed_total`, `job_duration_seconds`, `job_queue_depth`	counter / histogram / gauge	type, outcome, queue, status
`outbox_lag_seconds`, `outbox_published_total`	gauge / counter	type
`realtime_connections`, `realtime_messages_total`	gauge / counter	transport (websocket, sse)
`analytics_events_total`	counter	project, result
Node.js defaults	various	event-loop lag, heap, GC, CPU

Postgres and Redis are covered by `postgres_exporter` and `redis_exporter`. External synthetic checks (readiness through the gateway, TLS expiry) come from `blackbox_exporter`.

Dashboards: Grafana's "Surette Nexus — Overview" (`deploy/observability/grafana/dashboards`) has traffic and errors, latency percentiles, slowest routes, database pool and transaction time, cache hit ratio, rate limiting, queue, outbox, job outcomes, circuit breakers, external latency, live connections and analytics ingestion.

Alerts (`deploy/observability/prometheus/alerts.yml`, 20 rules; each links to a section of runbooks.md):

Severity → route	Alerts
critical → on-call (PagerDuty/Opsgenie webhook) + chat	service down, worker missing, readiness failing externally, error rate > 5 %, Postgres down, outbox stalled > 5 min
warning → chat + email	error rate > 1 %, p95 > 1.5 s, TLS < 14 days, database pool waiting, slow transactions, Postgres connections > 80 %, replica lag > 30 s, Redis down, dead-lettered jobs, queue > 500, outbox lag > 60 s, circuit open 5 min, rate-limit spike, analytics silent 3 daytime hours

Alertmanager groups by alert, repeats critical alerts hourly, and inhibits symptom alerts while the underlying service or database is down.

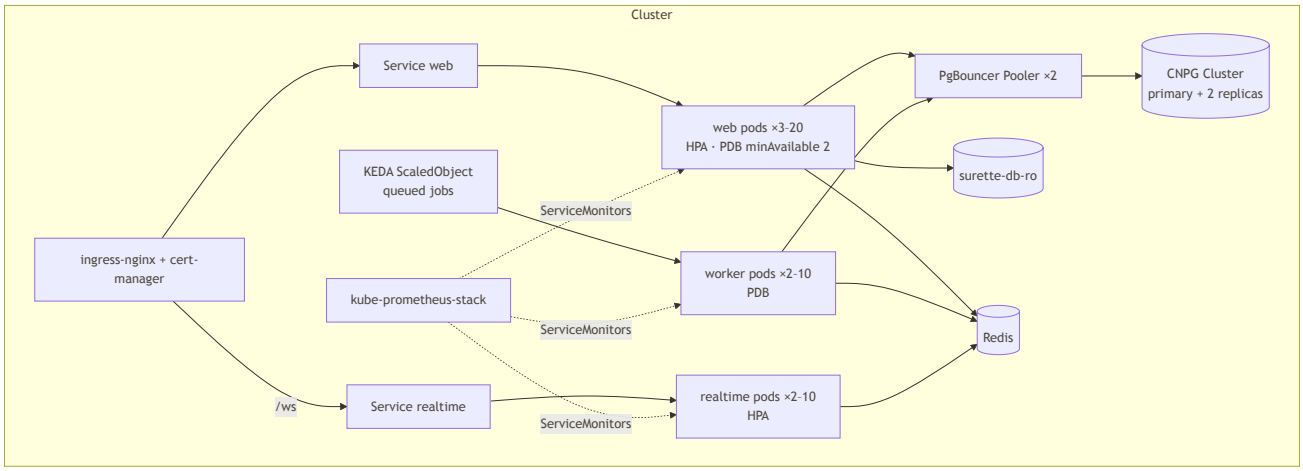
Tracing: not implemented yet. Request ids already correlate logs across gateway, web and worker. The next step is OpenTelemetry (`@opentelemetry/sdk-node`) exporting to Grafana Tempo.

14. Deployment

One image, three roles (`Dockerfile`): a multi-stage build, a non-root user (uid 10001), and tini as PID 1. The command picks the role: `npm start` (web), `npm run worker:prod` or `npm run realtime:prod`. Migrations run as a one-off (`npm run db:migrate:prod`).

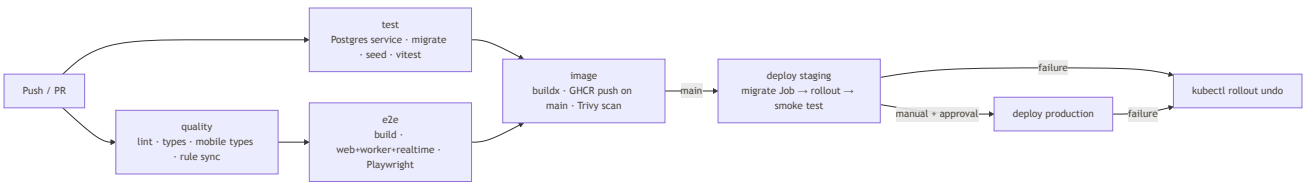
Docker Compose (`docker-compose.yml`): Postgres, Redis, a migrate one-shot, web ×2, worker, realtime and the nginx gateway on `:8080`. Add `--profile observability` for Prometheus, Alertmanager, Grafana (`:3001`), Loki, Promtail and the exporters, and `--profile ai` for Ollama.

Kubernetes (`deploy/k8s`, kustomize):



- Zero-downtime rollouts (`maxUnavailable: 0`, startup, readiness and liveness probes, `preStop` drain).
- Pods spread across zones; PodDisruptionBudgets keep capacity during node maintenance.
- Pod Security "restricted": non-root, read-only root filesystem, all capabilities dropped, seccomp.
- NetworkPolicies: default-deny ingress, with only the ingress controller allowed to web/realtime and only Prometheus allowed to metrics ports.
- Secrets come from a Secret managed by External Secrets / Sealed Secrets; there are none in git.
- Overlays: **staging** (1 replica, single-instance DB) and **production** (data layer, KEDA, monitoring).

15. CI/CD



Files: `.github/workflows/ci.yml` and `.github/workflows/deploy.yml`. The production deploy needs a person to choose the image and pass the GitHub Environment approval. Required secrets:

`KUBECONFIG` per environment; GHCR uses the built-in `GITHUB_TOKEN`.

Bugs found by actually running the stack (all fixed)

Found by	Bug	Fix
Docker build	Lock file generated on macOS was missing Linux packages, so `npm ci` failed in Docker and would have failed in CI	Lock regenerated in the Linux build image; `scripts/refresh-lockfile.sh` for future changes
Docker run	Upload folder owned by root while the app runs as a non-root user: photo and document uploads would fail	Image creates `/data/storage` owned by the app user
Docker run	The image health check probed the web port in every role, so worker and realtime were "unhealthy"	Role-aware health check (`HEALTH_PORT`, `HEALTH_PATH`)
Docker run	Scripts never exited when Redis was configured (open connection)	Scripts and the worker close the connection
Valkey tests	Redis client rejected commands while connecting, so the first requests after a start silently fell back	Commands wait for the connection, capped at 1 s each
Gateway tests	nginx resolved container addresses once; after a redeploy it sent traffic to addresses that no longer existed (502)	`resolver` + `resolve` on upstreams, re-resolved every 10 s
Gateway tests	Security headers sent twice (gateway and app)	Headers come only from the app
Gateway tests	Micro-cached responses replayed a previous request's `X-Request-Id`	Gateway hides the app's copy and sets its own
Accessibility (newer axe)	Logo link and "Ask ops" button text didn't match their accessible names	Names include the visible text
First GitHub CI runs	Route types missing on a fresh checkout; mobile lock file broken on Linux; image names must be lowercase; Trivy found 3 fixable critical Perl CVEs in the base image	`next typegen` before type-check; Linux-generated locks; lowercase `ghcr.io` names; `apt-get upgrade` in the image

16. Targets and capacity

Targets (SLOs) to hold ourselves to. These are **goals, not measurements**:

SLO	Target
Availability of public pages and read APIs	99.9 % monthly
p95 latency, read APIs (cache hit)	< 100 ms
p95 latency, write APIs	< 500 ms

SLO	Target
Live update delivery (commit → browser)	< 2 s
Webhook first attempt after commit	< 10 s

Measured (one production-build instance on a laptop, local Postgres, in-process cache, 25 concurrent clients; `npm run test:load`): cached listings API about **4,000 req/s at 13 ms p99**, market API 1,100 req/s at 43 ms p99, server-rendered home page 183 req/s at 205 ms p99, readiness 5,000 req/s, with zero errors. Full table in testing.md. The expected peak for a regional brokerage is in the tens of requests per second, so one instance has large headroom; three replicas are for availability, not load. Re-run against staging hardware before relying on these numbers in production. The first bottleneck under growth will be Postgres connections (PgBouncer), then analytics write volume (partitioning, §11).

Lesson from the load test: readiness originally failed whenever requests were queuing for a database connection. Under a spike, every pod would have reported "not ready" at once and been removed from the load balancer: a self-inflicted outage. Readiness now reflects only "can this pod serve at all" (database reachable, migrations applied). Pool pressure is a metric and an alert, and readiness results are cached for 2 s.

Verification status

Item	Status
Rate limiter (Postgres store), idempotency keys, job queue (SKIP LOCKED, retries, dead letter, dedupe), outbox (rollback safety, single fan-out with concurrent relayers), circuit breaker states and timeouts	Automated tests (`web/tests/integration/`, `web/tests/unit/circuit-breaker.test.ts`)
Health/readiness, metrics endpoint, request ids, idempotent replay over HTTP, 429 + Retry-After, WebSocket live update + cross-site rejection	Browser/HTTP tests (`web/e2e/platform/`)
Worker and realtime processes boot, schedule, relay and serve health/metrics	Run locally
Production `next build`	Passes
Alert rules in sync between Compose and Kubernetes, runbook for every alert, probes on every Deployment, gateway routing rules	Static tests (`web/tests/config/deploy-config.test.ts`); all YAML parses
Redis / Valkey backend (rate limits, shared cache, tag invalidation) and Redis outage fallback	Integration tests against local Valkey (`tests/integration/redis*.test.ts`)
Docker image + Docker Compose stack behind the nginx gateway	Built and run locally (Colima); site, accessibility, health, WebSocket and security tests pass through the gateway

Item	Status
Alert rules / Prometheus config / Alertmanager routing	`promtool check rules` (20 rules), `promtool check config`, `amtool check-config` pass
Kubernetes manifests (staging + production overlays, incl. CloudNativePG, KEDA, Prometheus Operator resources)	Rendered with `kubectl kustomize`; all 54 resources valid with `kubeconform -strict`
GitHub Actions CI (quality, tests on Postgres, browser tests, image build + Trivy scan, push to GHCR)	**Passing on GitHub**
Real Kubernetes cluster, Prometheus/Grafana/Loki containers, deploy workflow	**Not run yet** (needs hosting)
Load/capacity	Measured locally with autocannon (§16); not yet on staging hardware
Government datasets (Census, BLS, FHFA)	Downloaded and verified (checksums, completeness, plausibility), live-format contract tests pass
Accessibility	axe-core WCAG 2.1 AA audits pass on 15 pages